Multiple Criteria Decision Analysis

Goal Programing

Introduction to Multiple Criteria Decision Analysis (MCDA)

- MCDA is a decision-making framework for problems with multiple, often conflicting objectives
- Unlike single criterions methods (e.g. maximizing profit), MCDA balances tradeoffs.
- **Applications:**
- Business strategy, resource allocation, public policy
  - Trade-offs between criteria)
- Goal Programming ➔ A key MCDA technique that minimizes deviations from set goals.

Goal Programming – Core Concept

- A linear programming variant to minimize deviations from multiple goals
- Goals: Desired targets (e.g., profit ≥ $100M, employment = 50)
- Deviations: Overachievement ($d+$) or underachievement ($d-$)
- Objective: Minimize weighted sum of deviations of these objective functions from their respective goals.

Three Possible Types of Goals

- A lower, one-sided goal sets a lower limit that we do not want to fall under (but exceeding the limit is fine).
  - Example: Save at least $500 every month
- An upper, one-sided goal sets an upper limit that we do not want to exceed (but falling under the limit is fine).
  - Example: Spend no more then $200 on entertainment
- A two-sided goal sets a specific target that we do not want to miss on either side.
  - Number of employment = 50

Goal Programing – Mathematical programing

- Goal programming problems can be categorized according to the type of mathematical programming model including linear programming, integer programming, nonlinear programming, etc.
- We only consider linear goal programing
- Non-preemptive goal programing ➔ all the goals are of roughly comparable importance
- Preemptive goal programing ➔ there is a hierarchy of priority levels for the goals

Prototype Example - The Dewright Company Problem

- **Scenario:**
- Dewright Company is replacing discontinued products with three new ones.
- Decision: Determine production mix for Products 1, 2, and 3.
- Factors: Long-run profit, workforce stability, capital investment.

The Dewright Company Problem (Cont…)

- **Profit:** At least $125M (lower, one-sided).
- **Employment:** Exactly 4,000 employees (two-sided).
- **Capital Investment:** No more than $55M (upper, one-sided).

Data of Dewright Co.

TABLE 16.8 Data for the Dewright Co. nonpreemptive goal programming problem

	Unit Contribution				
	Product:				
Factor	**1**	**2**	**3**	**Goal (Units)**	**Penalty Weight**
Long-run profit	12	9	15	$\geq$ 125 (millions of dollars)	5
Employment level	5	3	4	$=$ 40 (hundreds of employees)	2(+), 4(−)
Capital investment	5	7	8	$\leq$ 55 (millions of dollars)	3

Columns Meaning in the Table

- **Goals**: The objectives (e.g., profit, employment, capital investment).
- **Product 1, 2, 3 (x_1, x_2, x_3)**: Coefficients showing how much each product contributes to each goal per unit produced.
- **Target Level**: The desired value for each goal.
- **Weight**: Penalty costs for deviating from the target (e.g., underachievement or overachievement).
- **Example:** Producing 1 unit of Product 1 adds $12 million to profit, 5 hundred employees to employment, and $5 million to capital investment.

The Dewright Company Problem

- Decision Variables: x_1,x_2, x_3 (production rates of Products 1, 2, 3).
- Goal
  - Profit: $12x_1 + 9x_2 + 15x_3 \geq 125$ (long-run profit, millions).
  - Employment: $5x_1 + 3x_2 + 4x_3 = 40$ (employment level, hundreds).
  - Investment: $5x_1 + 7x_2 + 8x_3 \leq 55$ (capital investment, millions).

Goal Programing Concept

- Objective: Minimize deviations from goals using penalty weights.
- Deviation Variables
  - d^- : Underachievement (amount below goal).
  - d^+ : Overachievement (amount above goal).
- General Form: Minimize Z = $\sum$(weight × deviation)
- z= $w_1 d_1^- + w_2 d_2^+$

Step 1. Formulating the Dewright Problem

1. Profit Goal: $12x_1 + 9x_2 + 15x_3 + d_1^- - d_1^+ = 125$

2. Employment Goal: $5x_1 + 3x_2 + 4x_3 + d_2^- - d_2^+ = 40$

3. Investment Goal: $5x_1 + 7x_2 + 8x_3 + d_3^- - d_3^+ = 55$

Non-negativity Constraints: $x_i \geq 0, d_i^- \geq 0, d_i^+ \geq 0$ *for* $i = 1{,}2{,}3$

- Equations aligned with arrows linking $d^-(under)\ and\ d^+(over)$
- We add $d^-(under)\ and\ d^+$(over) to each goal to measure deviations.
- For profit, we want $d_1^+ = 0$ (no overachievement penalty);…means more profit is acceptable.
- for employment, both deviations matter;
- for investment, d_3^- (no underachievement penalty)….means less investment is fine

Step 2: Define the Objective Function

- 5: Under profit goal.
- 2: Over employment goal.
- 4: Under employment goal.
- 3: Over investment goal.

$$\text{Minimize: } Z = 5d_1^- + 2d_2^+ + 4d_2^- + 3d_3^+$$

- The objective Z sums the weighted deviations. Higher weights (e.g., 5 for profit underachievement) show greater priority

A random mix

Let's try: $x_1 = 5, x_2 = 0, x_3 = 3.75$

Check Goals:

1. Profit: $12(5) + 9(0) + 15(3.75) = 60 + 56.25 = 116.25$
   - Under by $125 - 116.25 = 8.75$, so $d_1^- = 8.75, d_1^+ = 0.$
2. Jobs: $5(5) + 3(0) + 4(3.75) = 25 + 15 = 40$
   - Exactly 40, so $d_2^- = 0, d_2^+ = 0.$
3. Investment: $5(5) + 7(0) + 8(3.75) = 25 + 30 = 55$
   - Exactly 55, so $d_3^- = 0, d_3^+ = 0.$

Penalty: $Z = 5(8.75) + 4(0) + 2(0) + 3(0) = 43.75$

Another mix

- Mix 1: Penalty = 43.75
- Mix 2: Penalty = 75

Let's try: $x_1 = 0$, $x_2 = 0$, $x_3 = 10$

Check Goals:

1. Profit: $12(0) + 9(0) + 15(10) = 150$
   - Over by $150 - 125 = 25$, so $d_1^- = 0$, $d_1^+ = 25$.
2. Jobs: $5(0) + 3(0) + 4(10) = 40$
   - Exactly 40, so $d_2^- = 0$, $d_2^+ = 0$.
3. Investment: $5(0) + 7(0) + 8(10) = 80$
   - Over by $80 - 55 = 25$, so $d_3^- = 0$, $d_3^+ = 25$.

Penalty: $Z = 5(0) + 4(0) + 2(0) + 3(25) = 75$

Comparison of Mix 1 AND Mix 2

- **Results:**
  - Profit: $116.25M (a bit low, but okay).
  - Jobs: 4,000 (perfect!).
  - Investment: $55M (perfect!).
- **Penalty:** 43.75 points (from missing profit).
- Sometimes you can’t get everything—you have to choose what’s most important!

Why Goal Programming

- **Good Stuff:**
  - Helps us handle many goals at once.
  - Lets us decide what’s most important (using penalties).
- **Tricky Stuff:**
  - Choosing penalties can be hard (why 5 for profit but 4 for jobs?).
  - Takes time to test lots of mixes by hand.
- **Real Life:** You can use this for planning your time, budget, or even a group project!

Key Points to remember

- Goal programming helps balance multiple goals.
- Steps:
  - Set your goals.
  - Decide penalties for missing them.
  - Test mixes to find the best balance.

A simple example

- A construction company is planning to build three types of rooms—small, medium, and large, within a new facility. Each room type has a specific size (in square feet) and construction cost (in dollars). The company has a total available area of 25,000 square feet and a budget of $1,000,000. Due to space and budget constraints, the company may not be able to meet all its desired goals for the number of rooms of each type. Therefore, the company will use goal programming to balance multiple objectives, prioritizing deviations from its targets based on their relative importance.

Data

Type	Small	medium	Large		
Size	400	750	1050		
cost	18000	30000	45150		
Construction	Small	medium	Large	Total size	Total Budget
Target Value	5	10	15	25000	1000000

Python Output

```
Optimal Solution:
Number of Small Rooms (x1): 20.0
Number of Medium Rooms (x2): 15.0
Number of Large Rooms (x3): 4.0

Deviations:
Small Rooms Underachievement (y1_minus): 0.0
Small Rooms Overachievement (y1_plus): 0.0
Medium Rooms Underachievement (y2_minus): 0.0
Medium Rooms Overachievement (y2_plus): 0.0
Large Rooms Underachievement (y3_minus): 6.0
Large Rooms Overachievement (y3_plus): 0.0

Total Weighted Deviation (Objective Value): 36.0

Total Area Used: 23450.0 square feet
Total Cost: $990600.0
```

Defining Goal Constrains

- Total Expansion

$$400X_1 + 750X_2 + 1{,}050X_3 + d_4^- - d_4^+ = 25{,}000$$

- Total Cost (in $1,000s)

$$18X_1 + 33X_2 + 45.15X_3 + d_5^- - d_5^+ = 1{,}000$$

where

$$d_i^-, d_i^+ \geq 0$$